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Data-Driven Innovation for Trustworthy AI

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Abstract
Global concerns over the trustworthiness of rapidly proliferating artificial intelligence (AI)-centric artifacts have led to generic institutional recommendations for trustworthy AI, which have yet to be operationalized and integrated with design and innovation processes. We leverage the double hump model of data-driven innovation to propose and illustrate diverse data-driven approaches for identifying and evaluating opportunities, and generating and evaluating concepts for trustworthy AI. These approaches are expected to operationalize the institutional recommendations of trustworthy AI. Building on existing frameworks for classifying and managing risks associated with AI, we advocate for an ontological basis for trustworthy AI to enable fine-grained, computational assessments of AI-centric artifacts, their domains, and the organizations that develop or manage them.

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Need for Trustworthy Artificial Intelligence

As artifacts such as products, services, and systems are rapidly becoming artificial intelligence (AI)-centric, there has been a global concern regarding the trustworthiness of the AI constituents of these artifacts and their development processes.[1] In this article, we present a perspective on how a model of data-driven innovation can guide the development of trustworthy AI and support the assessment of existing AI-centric artifacts for their trustworthiness. We define an AI-centric artifact as,

> A technical or a socio-technical system, whose primary functions are shaped or enabled by AI technologies, comprising physical, digital, social and/or natural components that combine, in part or as a whole, to exhibit behaviors distinguishable from those of equivalent systems, which are not directly shaped or enabled by AI technologies.

For example, let us consider a headshot photo booth run by a photographer who, with the aid of various tools, performs the intended functions such as capturing photos, removing backgrounds, editing color tones, cropping, and resizing. In the AI-centric version of the same service, there are various AI-powered apps, like remove.bg[2] and photoAiD,[3] that perform many of these functions themselves upon uploading a photo or capturing one through a smartphone. The physical and digital components of these AI-centric artifacts are quite distinct from those of the photo booths bereft of AI support.[4]

In the example above, AI-centric photography services may not always be *robust* and *secure* enough to perform the desired editing functions *reliably*. As users upload their personal photos, their *privacy* can be compromised. In the event of system failure or privacy breach, it is unclear whether the app is *accountable* and has a *transparent* workflow for *auditability*. Moreover, it is unknown whether the app performs *fair* and *unbiased* editing for all kinds of users (e.g., gender and skin tone), thereby ensuring social well-being. These are part of a broader range of concerns that have led to the recent evolution of the field of research called Trustworthy AI.

Regarding the trustworthiness of rapidly evolving AI, Bhaskar Chakravorti lists 12 key concerns: disinformation, safety and security, the black box problem, ethical concerns, bias, instability, hallucinations in LLMs, unknown unknowns, potential job losses and social inequalities, environmental impact, industry concentration, and state overreach.[5] The concerns about trustworthiness involve not only the AI-centric artifacts but also the development processes and usage scenarios. A typical AI development process involves the use of unstructured (e.g., text documents) and labelled data (e.g., labelled images) to train models that perform tasks (e.g., image classification, and text generation) that translate to various applications.

- At the data level, concerns are tied to the origins, fairness, bias, and authenticity, among others. The CalGang dataset, which is used for crime prediction, has been found to be biased and unfair.[6] Similar concerns have also been raised against datasets used for mortgage approval, recruitment, and other purposes.
- At the model level, there is a lack of explainability because of architectures that are rapidly increasing in complexity. Radiologists have expressed

1 Bjorn Kleizen et al., "Do Citizens Trust Trustworthy Artificial Intelligence? Experimental Evidence on the Limits of Ethical AI Measures in Government," *Government Information Quarterly* 40, no. 4 (2023): article no. 101834, https://doi.org/10.1016/j.giq.2023.101834.
2 Removebg, see https://www.remove.bg/.
3 photoAiD, see https://photoaid.com/.
4 To add more, by AI-centric artifacts, we do not strictly consider those artifacts where AI-features are floated as extra supplements, which do not affect the primary functions of the product or service, e.g., virtual try-on features in eyewear shopping apps.
5 Bhaskar Chakravorti, "AI's Trust Problem: Twelve Persistent Risks of AI that Are Driving Skepticism," *Harvard Business Review*, May 3, 2024, https://hbr.org/2024/05/ais-trust-problem.
6 California Department of Justice, https://www.oag.ca.gov/sites/all/files/agweb/pdfs/bciis/ch7-6-isor.pdf.

concerns regarding the "black-box" nature of survival and prognosis predictions made using medical images.[7]

- At the application level, privacy and security concerns arose due to events such as the Equifax data breach.[8] Recently, the Federal Trade Commission (FTC) reported that consumers are less willing to support AI training through their text and voice opinions.[9]
- At the societal level, California nurses protested the use of AI-centric medical equipment, although it reportedly saved over 500 lives in 21 hospitals.[10] *Politico* reported how forged images, deepfake videos, and false robocalls negatively influence elections in various countries, including Britain, the USA, and Poland.[11]

The advent of AI has therefore brought trustworthiness that is currently not theorized and thus not operationalized enough to assess AI-centric artifacts. While literature on trustworthy AI is still at an early stage, several recommendations have been put forward by global initiatives (see next section). Upon reviewing these, we explain in the remainder of this article how models of design and innovation processes could guide the development of Trustworthy AI. We also briefly explain how existing AI-centric artifacts could be assessed for trustworthiness.

Generic Recommendations for Trustworthy AI

The European Commission designated a High-Level Expert Group on AI (HLEG-AI) in 2019 to develop ethics guidelines[12] for trustworthy AI. In 2020, these guidelines led to the development of an assessment checklist[13] for trustworthy AI. As stated in Table 1, other sets of recommendations have been put forward by the National Institute of Standards and Technology (NIST),[14] UNESCO,[15] Telefónica,[16] IBM,[17] and Deloitte.[18] The National Information Standards Organization (NISO) and the Association for Computing

7 Aimilia Gastounioti and Despina Kontos, "Is It Time to Get Rid of Black Boxes and Cultivate Trust in AI?," *Radiology: Artificial Intelligence* 2, no. 3 (2020): e200088, https://doi.org/10.1148/ryai.2020200088.

8 "Equifax Bata Breach," epic.org, accessed August 8, 2025, https://archive.epic.org/privacy/data-breach/equifax/.

9 Simon Fondrie-Teitler and Amritha Jayanti, "Consumers Are Voicing Concerns about AI," Federal Trade Commission, October 3, 2023, https://www.ftc.gov/policy/advocacy-research/tech-at-ftc/2023/10/consumers-are-voicing-concerns-about-ai#foot2.

10 Andrea Fox, "Nurses Have a Deep Distrust of AI — But Transparency and Training Could Help," *Healthcare IT News*, April 30, 2024," https://www.healthcareitnews.com/news/nurses-have-deep-distrust-ai-transparency-and-training-could-help.

11 Center for Digital Trust and EPFL, "Deepfakes, Distrust and Disinformation: The Impact of AI on Elections and Public Perception," Conference, October 1, 2024, SwissTech Convention Center, EPFL, https://c4dt.epfl.ch/c4dt-conference-on-disinformation-elections-and-ai/.

12 European Commission, "Ethics Guidelines for Trustworthy AI," April 8, 2019, https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai.

13 European Commission, "Assessment List for Trustworthy Artificial Intelligence (ALTAI) for Self-Assessment," July 17, 2020, https://digital-strategy.ec.europa.eu/en/library/assessment-list-trustworthy-artificial-intelligence-altai-self-assessment.

Table 1 **Generic Recommendations for Trustworthy AI put forward by global institutions.**

Initiative	Recommendations — Requirements, Criteria, or Principles
HLEG-AI	human agency & oversight, technical robustness & safety, privacy & data governance, transparency, diversity, non-discrimination & fairness, societal & environmental well-being, accountability
NIST	validity & reliability, safety, security & resiliency, accountability & transparency, explainability & interpretability, privacy, fairness with mitigation of harmful bias
UNESCO	proportionality & Do No Harm, safety & security, right to privacy & fata protection, multi-stakeholder and adaptive governance & collaboration, responsibility & accountability, transparency & explainability, human oversight & determination, sustainability, awareness & literacy, fairness & non-discrimination
Telefónica	fair AI, transparent and explainable AI, human-centric AI, privacy and security by design, working with partners and third parties
IBM	explainable, fair, robust, private, transparent
Deloitte	transparent & explainable, fair & impartial, robust & reliable, respectful of privacy, safe & secure, responsible & accountable

Machinery (ACM) recommend that AI development processes be repeatable, replicable, and reproducible.[19] While these recommendations also apply to responsible AI, trustworthy AI exclusively requires auditability, accountability, and liability.[20] These recommendations led to various impact assessment frameworks, including human rights (see the World Bank[21]), ethics (see UNESCO[22]), and privacy (see GDPR[23]).

Various scholarly contributions extend the recommendations in Table 1.[24] Bo Li et al. outline an actionable framework for trustworthy AI, recalling all recommendations and key players such as practitioners and societal stakeholders.[25] Qinghua Lu et al. present a catalogue for responsible AI—intended to be followed throughout the engineering lifecycle and governance.[26] In addition to fairness, explainability, accountability, and privacy, Kaur et al. include "acceptance" to ensure AI systems have empathy and morality.[27] Charles Radclyffe et al. suggest that the recommendations are assessed in tandem with fundamental rights such as non-discrimination, child protection, data protection, and freedom.[28] Li Zhang posits that the design of AI systems and the use of AI for designing are currently affirmative, while suggesting social and ethical design that upholds human values, such as collaboration, inclusiveness, and sustainability, and possibly confronts disruptive technologies that control human behavior through biological and cognitive means.[29]

Davinder Kaur et al. discuss the involvement of humans during AI development, specifically, before-the-loop (e.g., policymakers, domain experts), in-the-loop (e.g., data scientists, developers), and over-the-loop (e.g., regulatory bodies).[30] Haochen Liu et al. categorize the key roles of human beings in AI development into: system designers, decision makers, system deployers, system auditors, and end users.[31] Kars Alfrink et al. propose a "contestable" scheme for public AI that facilitates five loops of information exchange among stakeholders such as citizens, policymakers, developers, and public AI systems, involving mechanisms such as explanations, interactive controls, intervention requests, monitoring, participatory policymaking, participatory system development, tools for scrutiny, annual assessments, differential treatment, input data revisions, proactive notifications, and pro-social behavior incentives.[32]

The current recommendations for trustworthy AI are early propositions that are quite generic and have yet to be operationalized. Algorithms, methods, and toolkits are being proposed to address individual recommendations for trustworthiness, for example, regularizing functions to mitigate biases in classification models.[33] However, the literature does not include process models or roadmaps for the development and assessment of AI-centric artifacts. In the following sections, we leverage the design and innovation process model called the double hump model of data-driven innovation[34] to explain how these processes and their underlying methods could help guide the development of trustworthy AI-centric artifacts.

The Double Hump Model of Data-Driven Innovation

Jianxi Luo proposes the double hump model of data-driven innovation (Figure 1),[35] which suggests that design and innovation processes comprise

14 Framework for trustworthy and responsible AI given by the National Institute of Standards and Technology (NIST), https://www.nist.gov/trustworthy-and-responsible-ai.

15 UNESCO, "Ethics of Artificial Intelligence: The Recommendation," accessed July 14, 2025, https://www.unesco.org/en/artificial-intelligence/recommendation-ethics.

16 Telefónica, "AI Principles of Telefónica," October 2018, https://www.telefonica.com/en/wp-content/uploads/sites/7/2021/11/principios-ai-eng-2018.pdf.

17 IBM Watson, "Trustworthy AI," accessed July 14, 2025, https://research.ibm.com/topics/trustworthy-ai.

18 Deloitte, "Trustworthy AITM Services," accessed July 14, 2025, https://www2.deloitte.com/us/en/pages/deloitte-analytics/solutions/ethics-of-ai-framework.html.

19 Natalia Díaz-Rodríguez et al., "Connecting the Dots in Trustworthy Artificial Intelligence: From AI Principles, Ethics, and Key Requirements to Responsible AI Systems and Regulation," *Information Fusion* 99 (November 2023): article no. 101896, p. 9, https://doi.org/10.1016/j.inffus.2023.101896.

20 Ibid., 14.

21 World Bank Group, "Human Rights Impact Assessments: A Review of the Literature, Differences with Other Forms of Assessments and Relevance for Development," (Working Paper no. 125557, 2013), http://documents.worldbank.org/curated/en/834611524474505865.

22 UNESCO, *Ethical Impact Assessment: A Tool of the Recommendation on the Ethics of Artificial Intelligence* (Paris: United Nations Educational, Scientific and Cultural Organization, 2023), https://doi.org/10.54678/YTSA7796.

23 "Privacy Impact Assessment," GDPR.eu, accessed July 14, 2025, https://gdpr-info.eu/issues/privacy-impact-assessment/.

24 Kazuma Kobayashi and Syed Bahauddin Alam, "Explainable, Interpretable, and Trustworthy AI for an Intelligent Digital Twin: A Case Study on Remaining Useful Life," *Engineering Applications of Artificial Intelligence* 129 (March 2024): article no. 107620, https://doi.org/10.1016/j.engappai.2023.107620; Bhanu Chander et al., "Toward Trustworthy Artificial Intelligence (TAI) in the Context of Explainability and Robustness," *ACM Computing Surveys* 57, no. 6 (2025): article no. 144, https://doi.org/10.1145/3675392.

25 Bo Li et al., "Trustworthy AI: From Principles to Practices," *ACM Computing Surveys* 55, no. 9 (2023): article no. 177, https://doi.org/10.1145/3555803.

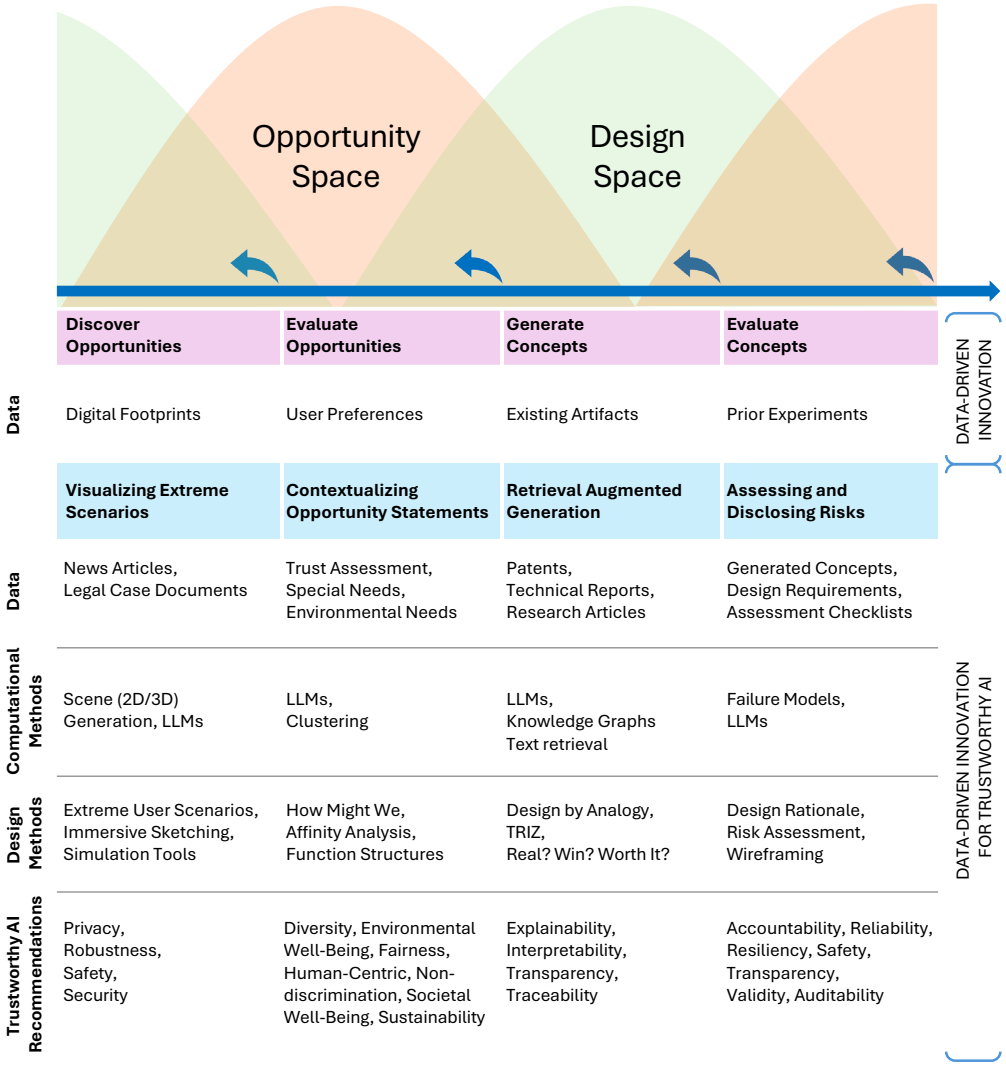

Figure 1
Data-Driven Innovation for Trustworthy AI. The early proposition of this model by Luo (2023) indicated that design and opportunity spaces are involved in four processes, suggesting various types of data that could be utilized in each process. In the extension of this model, as we propose in this article, we suggest specific data, computational, and design methods that can be used in each process. We also indicate the recommendations of trustworthy AI that can be addressed in each process. © 2025 L. Siddharth and Jianxi Luo.

discovering and evaluating opportunities that are capitalized by generating and evaluating concepts. The double hump model is based on the foundational theories of design creativity and innovation and integrates several research contributions that transformed design literature in the last decade—particularly in response to the growth of big data, complex systems, graph methods, deep learning, and language models.

Pioneering technology exploration in engineering design literature, Jianxi Luo et al. construct a map of patent classes, namely InnoGPS, to enable positioning, targeting, and navigating the total technology space in pursuit of technology and design opportunities[36] (*Discover Opportunities*). Bowen Yan and Jianxi Luo filter technology maps to identify the well-connected pairs of technology classes and the diversification paths of inventors and firms,[37] as demonstrated in case studies of prolific inventors and renowned firms such as Google (*Discover & Evaluate Opportunities*).

Upon filtering the technology classes tied to hybrid electric vehicles, Binyang Song et al. visualize a technology map explaining trajectories of technology expansion[38] (*Discover & Evaluate Opportunities*). Adopting various approaches to patent retrieval, such as keyword-based, citation-based, and inventor-based, Binyang Song and Jianxi Luo retrieve patents representing spherical rolling robots to provide inspirational ideas for concept generation[39] (*Generate Concepts*). In a related study on spherical rolling robots, V. Srinivasan et al. found that the retrieved patents with higher analogical distance from the design problem promote novelty and quality of the concepts generated[40] (*Evaluate Concepts*).

The above contributions leverage the existing structure of patent databases and adopt data- and graph-processing methods to explain and enhance design innovation. Following contributions adopt advanced natural language processing (NLP) and machine learning techniques to curate large amounts of data and drive design innovation processes. Upon crowdsourcing several concepts regarding future transportation systems, Yuejun He et al. perform core-periphery analysis on a semantic network of linguistic terms to identify the core concepts[41] (*Evaluate Concepts*). Serhad Sarica et al. obtain high-fidelity representations of technical terms in patent titles and abstracts through word-embedding models.[42] They utilize these representations to form semantic networks of targeted technology domains to aid in generating novel concepts of flying cars[43] (*Generate & Evaluate Concepts*).

Some recent contributions methodologically advance design literature through the adoption of LLM-centered applications for design and innovation processes. Qihao Zhu and Jianxi Luo fine-tune GPT-2 and GPT-3 models using patent titles and descriptions of Red Dot Award entries to generate concepts in response to target domains and problem statements[44] (*Generate Concepts*). To support idea generation inspired by nature, Zhu and Luo fine-tune GPT-3 using Ask Nature case descriptions to generate concepts in response to applications, benefits, and problem statements[45] (*Generate Concepts*). In both contributions, they generate flying car concepts and assess these using Word Mover's Distance (WMD), a proxy measure of novelty, to guide GPT models toward more novel solutions (*Evaluate Concepts*). Demonstrated with a detailed case study on battery thermal management, Shuo Jiang et al. use GPT-4 to interpret TRIZ contradictions from a design problem and then retrieve inventive principles for generating concepts[46] (*Generate Concepts*).

The above scholarly works contribute to theory and practice or data-driven processes integrated into the double hump model, as depicted in Figure 1. The double hump model partly adopts the structure of the widely utilized double-diamond representation (discover-define-develop-deliver) of design and innovation processes, as put forward by the UK Design Council.[47] This representation scheme has been instrumental in global design research, practice, and education, leading to some recent design outcomes such as traditional crafts,[48] outpatient services,[49] electric micro-vehicles,[50] and sex education apps.[51]

The double hump model alludes to the data that can act as a driver, not merely as a fuel, for design and innovation processes. The data used

26 Qinghua Lu et al., "Responsible AI Pattern Catalogue: A Collection of Best Practices for AI Governance and Engineering," *ACM Computing Surveys* 56, no. 7 (2024): article no. 173, https://doi.org/10.1145/3626234.

27 Davinder Kaur et al., "Trustworthy Artificial Intelligence: A Review," *ACM Computing Surveys* 55, no. 2 (2023): article no. 39, p. 23, https://doi.org/10.1145/3491209.

28 Charles Radclyffe et al., "The Assessment List for Trustworthy Artificial Intelligence: A Review and Recommendations," *Frontiers in Artificial Intelligence* 6 (March 9, 2023): article no. 1020592, pp. 4–5, https://doi.org/10.3389/frai.2023.1020592.

29 Li Zhang, "The Ethical Turn of Emerging Design Practices," *She Ji: The Journal of Design, Economics, and Innovation* 9, no. 3 (2023): 315–16, https://doi.org/10.1016/j.sheji.2023.09.002.

30 Kaur et al., "Trustworthy Artificial Intelligence," 8.

31 Haochen Liu et al., "Trustworthy AI: A Computational Perspective," *ACM Transactions on Intelligent Systems and Technology* 14, no. 1 (2023): article no. 4, p. 37, https://doi.org/10.1145/3546872.

32 Kars Alfrink et al., "Envisioning Contestability Loops: Evaluating the Agonistic Arena as a Generative Metaphor for Public AI," *She Ji: The Journal of Design, Economics, and Innovation* 10, no. 1 (2024): 62–64, https://doi.org/10.1016/j.sheji.2024.03.003.

33 Anoop Krishnan and Ajita Rattani, "A Novel Approach for Bias Mitigation of Gender Classification Algorithms Using Consistency Regularization," *Image and Vision Computing* 137 (September 2023): article no. 104793, https://doi.org/10.1016/j.imavis.2023.104793.

34 Jianxi Luo, "Data-Driven Innovation: What Is It?," *IEEE Transactions on Engineering Management* 70, no. 2 (2023): 784–90, https://doi.org/10.1109/TEM.2022.3145231.

35 Ibid.

36 Jianxi Luo et al., "InnoGPS for Data-Driven Exploration of Design Opportunities and Directions: The Case of Google Driverless Car Project," *Journal of Mechanical Design* 139, no. 11 (2017): article no. 111416, https://doi.org/10.1115/1.4037680.

37 Bowen Yan and Jianxi Luo, "Filtering Patent Maps for Visualization of Diversification Paths of Inventors and Organizations," *Journal of The Association for Information Science and Technology* 68, no. 6 (2017): 1551–63, https://doi.org/10.1002/asi.23780.

38 Binyang Song et al., "Overlay Technology Space Map for Analyzing Design Knowledge Base of a Technology Domain: The Case of Hybrid Electric Vehicles," *Research*

in Engineering Design 30, no. 3 (March 2019): 405–23, https://doi.org/10.1007/s00163-019-00312-w.

39 Binyang Song and Jianxi Luo, "Mining Patent Precedents for Data-Driven Design: The Case of Spherical Rolling Robots," *Journal of Mechanical Design* 139, no. 11 (2017): article no. 111420, https://doi.org/10.1115/1.4037613.

40 V. Srinivasan et al., "Does Analogical Distance Affect Performance of Ideation?," *Journal of Mechanical Design* 140, no. 7 (2018): article no. 071101, https://doi.org/10.1115/1.4040165.

41 Yuejun He et al., "Mining and Representing the Concept Space of Existing Ideas for Directed Ideation," *Journal of Mechanical Design* 141, no. 12 (2019): article no. 121101, https://doi.org/10.1115/1.4044399.

42 Serhad Sarica et al., "TechNet: Technology Semantic Network Based on Patent Data," *Expert Systems with Applications* 142 (March 2020): article no. 112995, https://doi.org/10.1016/j.eswa.2019.112995.

43 Serhad Sarica et al., "Idea Generation with Technology Semantic Network," *AI EDAM* 35, no. 3 (2021): 265–83, https://doi.org/10.1017/S0890060421000020.

44 Qihao Zhu and Jianxi Luo, "Generative Transformers for Design Concept Generation," *Journal of Computing and Information Science in Engineering* 23, no. 4 (2023): article no. 041003, https://doi.org/10.1115/1.4056220.

45 Qihao Zhu et al., "Biologically Inspired Design Concept Generation Using Generative Pre-trained Transformers," *Journal of Mechanical Design* 145, no. 4 (2023): article no. 041409, https://doi.org/10.1115/1.4056598.

46 Shuo Jiang et al., "AutoTRIZ: Automating Engineering Innovation with TRIZ and Large Language Models," *Advanced Engineering Informatics* 65, part C (May 2025): article no. 103312, https://doi.org/10.1016/j.aei.2025.103312.

47 Double Diamond representation of design and innovation processes, https://www.designcouncil.org.uk/our-resources/the-double-diamond/.

48 I-Chia Tsai and Yu-Yun Chen, "Survival Strategies for the Transformation of Traditional Japanese Crafts Based on the Double Diamond Model," *Creative Industries Journal* (April 5, 2024): online, https://doi.org/10.1080/17510694.2024.2329786.

49 Zhiwei Zhou et al., "The Design of Outpatient Services in Children's Hospitals Based on the Double Diamond Model," in *Digital Human Modeling and Applications in Health, Safety, Ergonomics and Risk Management. AI, Product and Service*, ed. Vincent G. Duffy (Cham: Springer, 2021), 182–93, https://doi.org/10.1007/978-3-030-77820-0_14.

to perform typical engineering operations (e.g., finite element method) and user studies (e.g., surveys, interviews) can only be considered as a fuel—hence Luo labeled it as "data-based innovation."[52] In contrast, "data-driven innovation" requires leveraging data that can enhance the creativity and knowledge that drives these processes. For instance, engineers can leverage design rationale from organizational history to prevent engineering faults,[53] synthesize the latest technologies through patent retrieval,[54] and even generate synthetic data using AI to perform trial experiments.

Alfrink et al. posit that AI technologies are being built by non-designers,[55] who may not necessarily follow the structured design and innovation processes as proposed in the literature. Although several recommendations for trustworthy AI have been put forward (as reviewed in the above section on Generic Recommendations for Trustworthy AI), they remain insufficiently operationalized. They are often addressed retrospectively during the documentation stages that follow artifact development. Integrating key contributions from design and innovation literature, the double hump model offers recommendations on the data sources and methods that can be leveraged during each process (see Figure 1) to facilitate data-driven innovation. In the following section—Data Driven Innovation for Trustworthy AI—we explain and illustrate how this model can offer a structured roadmap for the development of trustworthy AI.

Data-Driven Innovation for Trustworthy AI

Discover Opportunities for Trustworthy AI

As shown in Figure 1, the first process of the double hump model is to discover design opportunities from relevant public or organizational data, like digital footprints from users and innovators.[56] As reviewed in the above section—The Double Hump Model of Data-Driven Innovation—prior research leverages technology maps from patent data to identify new design opportunities. The purpose of this section is to emphasize how designers, engineers, and other stakeholders can take such a data-driven approach to discover focused opportunities that can transition current AI-centric artifacts to be more trustworthy.

In the trustworthy AI discourse, scholars point toward critical incidents that occurred with artifacts from different domains. Amika Singh and Munindar Singh refer to the 1972 explosion of the Ford Pinto due to the incorrect placement of the gas tank.[57] Kaur et al. recall the critical issues in the electric car software of Volkswagen.[58] For artifacts to be trustworthy, it is necessary to handle such critical situations that enable typical performance and, potentially, inform new performance indicators.[59] For this purpose, as indicated in Figure 1, existing design methods recommend envisioning extreme scenarios and performing immersive sketching to generate solutions that tackle such scenarios.[60]

From the perspective of data-driven innovation, we propose to enhance these methods by retrieving the knowledge of critical situations from news articles and legal case documents, as illustrated in Table 2.[61] To visualize extreme scenarios from these situations, we propose adopting 2D or 3D

Table 2 **Sample of news articles retrieved using the *New York Times* API.**

Abstract	URL	Date
"Mainstream companies have concerns about the lineage of the data that powers A.I. applications. An industry group is addressing that challenge."	https://www.nytimes.com/2023/11/30/business/ai-data-standards.html	2023–11–30
"The tech company's plan is similar to one announced two days earlier by Meta, another Silicon Valley giant."	https://www.nytimes.com/2024/02/08/business/media/google-ai.html	2024–02–08
"Beware Silicon Valley's latest trend: fake transparency."	https://www.nytimes.com/2021/08/08/opinion/tiktok-facebook-youtube-transparency.html	2021–08–08

Note: These news articles are examples of AI-related incidents that can inform AI development processes across domains, helping to address critical performance concerns such as robustness, privacy, security.

scene generation models[62] that can output scenarios in response to text prompts.

It is not common in design and innovation literature to recommend sources such as news articles and legal case documents to discover opportunities. In engineering design practice, problems are typically identified using the design rationale documents that comprise issues, decisions, and explanations either from organizational history or domain-specific reports.[63] As artifacts are increasingly becoming AI-centric, many industries may not have sufficient AI-centric design history to learn and improve from the past.

In addition, even those AI-centric artifacts from different domains likely share similar infrastructure (models, hardware, hosting services) for AI implementation. For example, Meta's LLaMA model powers document review and code generation in Goldman Sachs,[64] and supports meeting summaries and message completion in Zoom.[65] In terms of hardware, LiDAR and infrared sensors are commonly used in computer vision applications across domains such as healthcare, robotics, and transportation. A critical failure in AI infrastructure in one domain can therefore offer useful lessons for others and thus provide new design opportunities.

In terms of trustworthiness, the primary issues related to AI-based artifacts concern robustness, privacy, security, and safety (indicated in Figure 1). Less robust AI models are prone to various forms of attack, such as those involving mislabeled training or test data — for example, introducing noise or modifying data entries to cause misidentification.[66] Liu et al. showcase examples where image classifiers fail to identify objects like pandas and stop signs.[67] Adversarial training has been recommended for defending against attacks on training data.[68] To address privacy-related issues, models could be trained with datasets without personal information labels or retrained with less private data, i.e., machine unlearning.[69] Additionally, Natalia Díaz-Rodríguez et al. suggest federated learning, homomorphic computing, and differential privacy.[70] In federated learning, Zhanpeng Yang et al. propose a blockchain-based learning architecture intended to prevent Byzantine attacks and model tampering.[71]

50 Sara Viviani et al., "An Interdisciplinary Double-Diamond Design Thinking Model for Urban Transport Product Innovation: A Design Framework for Innovation Combining Mixed Methods for Developing the Electric Microvehicle 'Leonardo Project,'" *Energies* 17, no. 23 (2024): article no. 5918, https://doi.org/10.3390/en17235918.

51 Clarissa Soehnchen et al., "Creating an Intercultural User-Centric Design for a Digital Sexual Health Education App for Young Women in Resource-Poor Regions of Kenya: Qualitative Self-Extended Double Diamond Model for Requirements Engineering Analysis," *JMIR Formative Research* 7 (2023): online, https://doi.org/10.2196/50304.

52 Luo, "Data-Driven Innovation," 784.

53 Rob Bracewell et al., "Capturing Design Rationale," *Computer-Aided Design* 41, no. 3 (2009): 173–86, https://doi.org/10.1016/j.cad.2008.10.005.

54 L. Siddharth et al., "Enhancing Patent Retrieval Using Text and Knowledge Graph Embeddings: A Technical Note," *Journal of Engineering Design* 33, no. 8-9 (2022): 670–83, https://doi.org/10.1080/09544828.2022.2144714.

55 Alfrink et al., "Envisioning Contestability Loops," 59.

56 Luo, "Data-Driven Innovation," 3.

57 Amika M. Singh and Munindar P. Singh, "Wasabi: A Conceptual Model for Trustworthy Artificial Intelligence," *Computer* 56, no. 2 (2023): 23, 25, https://doi.org/10.1109/MC.2022.3212022.

58 Kaur et al., "Trustworthy Artificial Intelligence," 18.

59 Singh and Singh, "Wasabi," 22.

60 Vincent Rieuf et al., "Emotional Activity in Early Immersive Design: Sketches and Moodboards in Virtual Reality," *Design Studies* 48 (January 2017): 43–75, https://doi.org/10.1016/j.destud.2016.11.001.

Evaluate Opportunities for Trustworthy AI

The second process of the double hump model is to evaluate and refine the opportunities thus discovered in the previous process (Figure 1). Luo recommends that opportunities be evaluated and refined in tandem with users' digital footprints, which include their opinions expressed on social media and e-commerce platforms.[72] As AI-centric products and services have only recently become mainstream, it is difficult to mine large amounts of opinions from a wide range of users (bottom-up approach) about mainstream, real-world use cases, excluding the exploratory usage of tech enthusiasts.

To address the lack of large amounts of product usage data, as indicated in Figure 1, we propose leveraging the trustworthiness assessment checklist (top-down approach) given by HLEG-AI and discussed in detail by Radclyffe et al., along with a human rights assessment.[73] To leverage the trustworthiness assessment checklist, we additionally propose generating "How might we" type questions that are commonly generated as part of design methods (as indicated in Figure 1). In Table 3, we demonstrate our proposition in the context of an AI-centric editorial assistant that can be used to assess and classify journal paper submissions and assist with several editorial tasks. It is critical that such an assistant exhibits less bias and makes explainable decisions. In this context, we utilize two standard assessment questions discussed by Radclyffe et al., regarding diversity and traceability.[74] We first suggest simplifying and contextualizing these questions into short opportunity statements as shown in Table 3. Additionally, we suggest using LLMs for these tasks — the statements in Table 3 were generated by GPT-4. As many opportunity statements are generated, as indicated in Figure 1, design methods such as affinity diagrams and function structures can be adopted to organize and discuss these, preferably leveraging NLP techniques along with Sentence Transformers for clustering.[75]

Adopting the above-stated approaches can operationalize the institutional recommendations for trustworthy AI, such as explainability, social,

61 The *New York Times* Article Search API, https://developer.nytimes.com/docs/articlesearch-product/1/overview.

62 Rafail Fridman et al., "SceneScape: Text-Driven Consistent Scene Generation," in *NIPS '23: Proceedings of the 37th International Conference on Neural Information Processing Systems*, ed. A. Oh et al. (New York: ACM, 2023), article no. 1734, https://dl.acm.org/doi/10.5555/3666122.3667856.

63 L. Siddharth et al., "Modeling and Structuring Design Rationale to Enable Knowledge Reuse," *Systems Engineering* 23, no. 3 (2020): 294–311, https://doi.org/10.1002/sys.21515.

64 "Goldman, Nomura Tap Meta Llama AI Models," Finextra, August 30, 2024, https://www.finextra.com/newsarticle/44650/goldman-nomura-tap-meta-llama-ai-models?utm_source=chatgpt.com.

65 Meta's report on the usage of their Llama model in Zoom, https://ai.meta.com/blog/llama-zoom-ai-companion/?utm_source=chatgpt.com.

66 Liu et al., "Trustworthy AI," 8, 12.

67 Ibid.

68 Lue Tao et al., "Better Safe Than Sorry: Preventing Delusive Adversaries with Adversarial Training," in *NIPS '21: Proceedings of the 35th International Conference on Neural Information Processing Systems*, ed. M Ranzato et al. (New York: ACM, 2021), article no. 1240, https://dl.acm.org/doi/10.5555/3540261.3541501.

69 Yinzhi Cao and Junfeng Yang, "Towards Making Systems Forget with Machine Unlearning," in *2015 IEEE Symposium on*

Table 3 **Illustrating opportunity statements derived from the standard assessment checklist.**

Standard Assessment Checklist	Abstract Opportunity Statements	Contextualized Opportunity Statements
"Did you consider diversity and repetitiveness of end-users and/ or subjects in the data?"	- How might we enhance diversity among end-users and subjects in our dataset? - How might we minimize the repetitiveness of end-users and subjects in our data? - How might we ensure inclusivity in our data regarding end-users and subjects?	- How might we include articles from all disciplines in our dataset? - How might we recruit editors of different levels of expertise? - How might we consider articles of all types and lengths?
"Can you trace back which data was used by the AI system to make a certain decision(s) or recommendation(s)?"	- How might we implement a system to track the data used for AI decisions or recommendations? - How might we enable easy retrieval of data utilized in AI decision-making? - How might we ensure transparency in accessing data behind AI decisions and recommendations?	- How might we identify the specific article portions that led to the classification? - How might we retrieve example articles that are relevant to the classification? - How might we visualize the features as a network?

Note: In the context of developing an AI-centric editorial assistant for journal editors.

and environmental well-being. Training and deployment of AI models incur direct and indirect environmental impacts.[76] For instance, generating 1,000 images with Stable Diffusion XL was found to be nearly as impactful as driving a car for 4.1 miles.[77] In terms of the impacts on social well-being, a study reports biased AI predictions in criminal case investigations and home loan approvals.[78] Computational solutions have been proposed in AI literature to address these concerns.

To mitigate environmental impacts at the model level, Tien-Ju Yang et al. propose pruning strategies for CNN models.[79] To mitigate model biases, Marc-Etienne Brunet et al. suggest removing a minor portion of a dataset to identify biases exhibited by word embedding models.[80] Black et al. introduce a Flip Test that produces a transparency report for classifiers, ranking all the features that are responsible for discriminating against protected groups of individuals.[81] Emily Kaur et al. review approaches such as balanced datasets, regularizing functions, weighted labeling, and other toolkits.[82] It is important to note that these strategies are often adopted at the cost of model performance.[83] While federated learning is widely proposed for model and data privacy, scholars acknowledge the lack of fairness and other responsible AI attributes in these approaches. Demonstrated using the Covid-19 X-ray dataset, Sin Kit Lo et al. propose a blockchain-based architecture with weighted fair data sampling to ensure fairness in federated learning.[84]

Some explainable AI solutions include issue-based explanations, counterfactual studies, and simplified model surrogates.[85] Meade Zolanvari et al. propose an explainable AI approach that uses factor analysis to transform input features into latent variables and the mutual information method to rank the variables from most to least important.[86] Kaur et al. suggest data labelling, exemplar justifications, what-if scenarios, rule-based approaches, and perturbation-based explanations.[87] Liu et al. illustrate how methods such as Decision Tree, GNN Explainer, Grad-CAM, and LIME could be used to explain and/or walk through how classification models make their predictions.[88] Although these solutions offer an immediate workaround, they are applied retrospectively—after AI models have been trained and integrated with AI-centric artifacts—without addressing trustworthiness concerns during earlier stages such as dataset creation or procurement. As a result, issues like bias may persist. We argue that these concerns should be injected much earlier into the design processes as opportunity statements, illustrated in Table 3.

Generate Concepts for Trustworthy AI

The third process of the double hump model is to generate solution concepts for the opportunities discovered and evaluated (Figure 1) in the previous processes. Luo proposes that generating concepts can be driven by repositories of existing artifacts, instead of relying on human expertise and serendipity in common design methods like brainstorming.[89] For an AI-centric artifact to be trustworthy, it is recommended that its underlying concepts are explainable, traceable, and transparent (as indicated in Figure 1). Hence, we propose to generate concepts through Retrieval-Augmented Generation (RAG) techniques that guide LLMs to generate traceable responses, thereby enabling their explainability and transparency.

Security and Privacy (San Jose, CA: IEEE, 2015), 463–80, https://doi.org/10.1109/SP.2015.35.

70 Díaz-Rodríguez et al., "Connecting the Dots," 10.

71 Zhanpeng Yang et al., "Trustworthy Federated Learning via Blockchain," *IEEE Internet of Things Journal* 10, no. 1 (2023): 92–109, https://doi.org/10.1109/JIOT.2022.3201117.

72 Luo, "Data-Driven Innovation," 787.

73 Radclyffe et al., "Assessment List."

74 Ibid., 6.

75 Code level implementation of Sentence Transformers commonly used to obtain text embeddings for downstream tasks such as clustering: https://huggingface.co/sentence-transformers.

76 Chakravorti, "AI's Trust Problem."

77 Sasha Luccioni et al., "Power Hungry Processing: Watts Driving the Cost of AI Deployment?," in *FAccT '24: Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (New York: ACM, 2024), 85–99, https://doi.org/10.1145/3630106.3658542.

78 Julia Dressel and Hany Farid, "The Accuracy, Fairness, and Limits of Predicting Recidivism," *Science Advances* 4, no. 1 (2018): online, https://doi.org/10.1126/sciadv.aao5580.

79 Tien-Ju Yang et al., "Designing Energy-Efficient Convolutional Neural Networks Using Energy-Aware Pruning," in *Proceedings of the 30th IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2017)* (Honolulu: IEEE, 2017), 6071–79, https://doi.org/10.1109/CVPR.2017.643.

80 Marc-Etienne Brunet et al., "Understanding the Origins of Bias in Word Embeddings," in *Proceedings of the 36th International Conference on Machine Learning*, ed. K Chaudhuri and R Salakhutdinov (Long Beach, CA: PMLR 97, 2019), 803–11, available at https://proceedings.mlr.press/v97/brunet19a.html.

81 Emily Black et al., "FlipTest: Fairness Testing via Optimal Transport," in *FAT* '20: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (New York: ACM, 2020), 111–21, https://doi.org/10.1145/3351095.3372845.

82 R. K. E. Bellamy et al., "AI Fairness 360: An Extensible Toolkit for Detecting and Mitigating Algorithmic Bias," *IBM Journal of Research and Development* 63, no. 4/5 (2019): 4:1–4:15, https://doi.org/10.1147/JRD.2019.2942287; Emmanouil Krasanakis et al., "Adaptive Sensitive Reweighting to Mitigate Bias in Fairness-Aware Classification," in *WWW '18:*

Proceedings of the 2018 World Wide WEB Conference (New York: ACM, 2018), 853–62, https://doi.org/10.1145/3178876.3186133; Kaur et al., "Trustworthy Artificial Intelligence."

83 Liu et al., "Trustworthy AI," 20.

84 Sin Kit Lo et al., "Toward Trustworthy AI: Blockchain-Based Architecture Design for Accountability and Fairness of Federated Learning Systems," *IEEE Internet of Things Journal* 10, no. 4 (2023): 3276–84, https://doi.org/10.1109/JIOT.2022.3144450.

85 Díaz-Rodríguez et al., "Connecting the Dots," 11; Melkamu Mersha et al., "Explainable Artificial Intelligence: A Survey of Needs, Techniques, Applications, and Future Direction," *Neurocomputing* 599 (September 2024): article no. 128111, https://doi.org/10.1016/j.neucom.2024.128111.

86 Meade Zolanvari et al., "Trust XAI: Model-Agnostic Explanations for AI with a Case Study on IIoT Security," *IEEE Internet of Things Journal* 10, no. 4 (2023): 2967–78, https://doi.org/10.1109/JIOT.2021.3122019.

87 Kaur et al., "Trustworthy Artificial Intelligence," 15, 16.

88 Liu et al., "Trustworthy AI," 23–26.

89 Luo, "Data-Driven Innovation," 787.

90 Zhu et al., "Biologically Inspired Design Concept Generation."

91 L. Siddharth and Jianxi Luo, "Retrieval Augmented Generation Using Engineering Design Knowledge," *Knowledge-Based Systems* 303 (November 2024): article no. 112410, https://doi.org/10.1016/j.knosys.2024.112410.

92 Díaz-Rodríguez et al., "Connecting the Dots," 10.

As reviewed in the above section, the recent research in data-driven innovation reports versions of GPT-3 that can generate novel and bioinspired solutions by learning inherent design principles from sources such as Red Dot Award, Ask Nature, and others.[90] In such concept-generation tools, GPT may generate a hypothetical concept description—for example, a marine robot shaped and behaving like a fish for managing aquatic life. While the generated concept may be novel, it is difficult to understand its feasibility without gathering sufficient scientific background and prior design concepts of equivalent marine robots. To address this issue of traceability, we demonstrate a RAG system where LLMs such as GPT are paired with large amounts of patent data and research articles.

To demonstrate a case example, we leverage our prior method[91] to populate knowledge graphs from 159 patents and 56 research articles related to carbon-capture systems. These knowledge graphs are explicit representations of design knowledge extracted from the technical documents by identifying facts at the sentence level. Unlike documents that are available in text format, knowledge graphs allow us to select and synthesize portions of knowledge from different sources. In addition, knowledge graphs are recommended for LLMs to generate domain-specific, technically rich, and source-traceable responses.[92]

In Table 4, we detail the process of retrieving the documents and extracting the knowledge graphs from these. Among 8.8 million patents granted by

Table 4 **Retrieving knowledge of carbon capture systems as of June 3, 2024.**

Step	Outcome
Patent Search (PatentsView, USPTO)	
Import granted patents	8,798,930 patents
Search for "carbon capture" in titles and abstracts	196 patents
Filter utility patents since 2014	159 patents
Article Search (Web of Science)	
Search for "carbon capture" in titles	7,686 papers
Filter articles since 2014	4,506 papers
Filter highly cited articles	56 papers
Natural Language Processing	
Combine results	215 documents
Clean and break paragraphs	1,233 sentences
Knowledge Retrieval	
Extract Facts	7,247 facts
Process Facts	7,219 facts
Remove duplicates within documents	6,841 facts
Amongst the top 50 frequent entities	242 facts
Get unique facts	144 facts
Most frequent relation between a pair of entities	120 facts

93 Siddharth and Luo, "Retrieval Augmented Generation."
94 Domain of carbon capture systems, https://carboncapture.vercel.app/.

USPTO as of June 3, 2024, we retrieved 159 patents of the last decade, including the term *carbon capture*, either in the titles or abstracts. We also searched the Web of Science for articles including the search term in the titles. As the search resulted in 7,686 papers, we only searched within titles and filtered 56 papers published in the last decade and recognized as highly cited papers. Overall, we retrieved 215 technical publications and acquired 1,233 sentences. We applied our prior method[93] to identify 7,247 facts from these sentences. After removing duplicates, we obtained 6,841 distinct facts from the knowledge graphs. From these, we select the most frequent 120 facts as visualized in Figure 2.[94]

Figure 2 serves as an explicit representation of generalizable knowledge of carbon-capture systems. Beneath Figure 2, we provide a description of the domain of carbon-capture systems, generated by GPT-4, when augmented with the selected 120 facts. The response by GPT-4 synthesizes individual facts into a coherent description of different aspects of design, including hierarchy, purposes, spatial or behavioral interactions, and transitions.

"Carbon capture systems (CCS) focus on the conversion of carbon dioxide (CO_2) from various sources such as air and flue gas into either more stable or

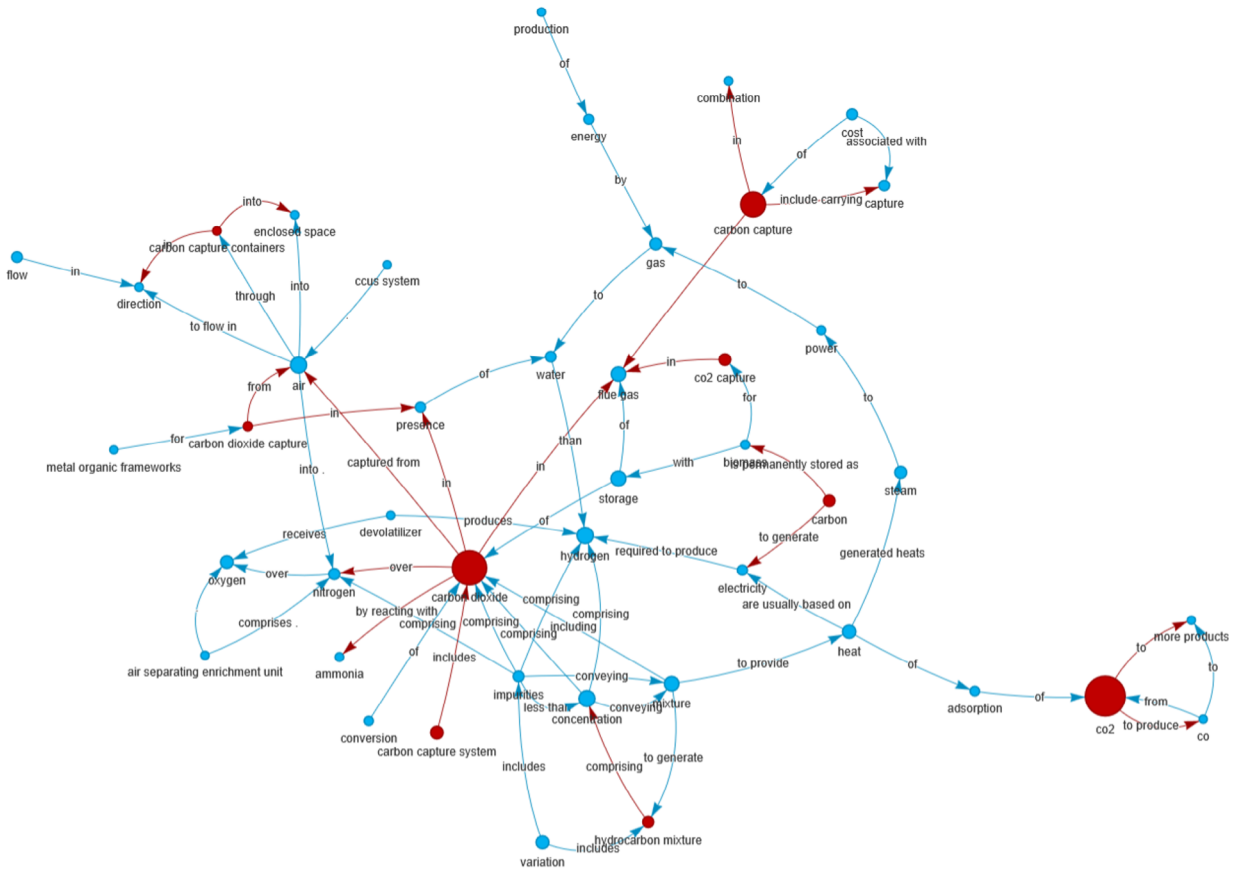

Figure 2
Representing generalizable knowledge of the domain of carbon capture using a portion of the 120 most frequent facts extracted from relevant patents and research articles. An interactive version of this knowledge graph, along with document references, can be accessed at https://carboncapture.vercel.app/.
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reusable forms. Critical to this process is the capturing of CO_2 using metal organic frameworks, followed by its storage in different mediums such as biomass or liquid forms. Systems often involve the use of adsorption techniques where CO_2 is captured in the presence of impurities. In compounds like flue gas, which also contains nitrogen, water, and varying concentrations of other gases, carbon dioxide can be captured and separated. The captured CO_2 might then react with chemicals like ammonia or be subjected to conditions in devices like bipolar membrane electrodialysis apparatus for further purification and utilization, for example in the generation of electricity through processes like carbon capture and utilization (CCU), where it is turned into carbon monoxide (CO) or synthesized into other products. Additionally, the longitudinal flow of these gases through systems such as carbon capture containers and the consumption of resources like heat, electricity, and biomass are also significant in the operational intricacies and cost determination of these systems."

The concepts indicated in the above-shown synthesis by GPT-4 can be identified in the interactive version[95] of the knowledge graph, which also includes references to the original document (patent or scientific article) for traceability. Given the opportunity statements generated in the previous process (Section: Evaluate Opportunities for Trustworthy AI), the above-demonstrated RAG activity can be adopted heuristically for retrieving specific knowledge of a domain and generating solution concepts. Such knowledge traceable concepts can be integrated into AI-centric artifacts, thereby improving their explainability and transparency.

95 Domain of carbon capture systems, https://carboncapture.vercel.app/.

Evaluate Concepts for Trustworthy AI

The final process of the double hump model is to evaluate solution concepts generated in the previous process (Figure 1). In Figure 3, we mention the evaluation criteria to mitigate risk and verify whether the generated concepts satisfy the requirements necessary for trustworthy AI. In the design

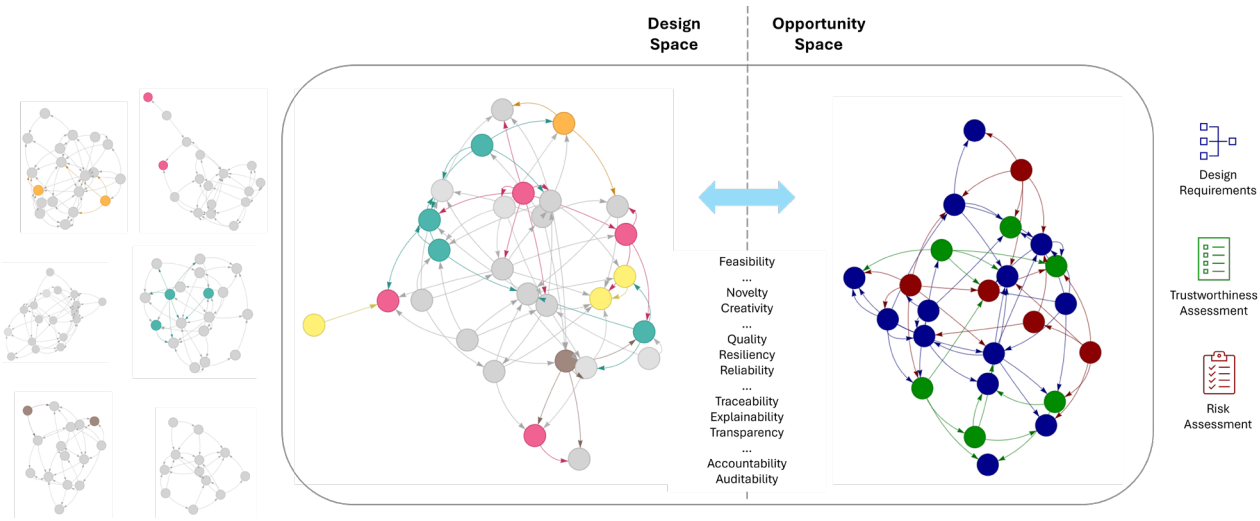

Figure 3
Evaluating concepts for trustworthy AI. The proposed scheme builds upon the knowledge graph-based synthesis of knowledge from existing artifacts to develop a solution concept that is evaluated against the design requirements, trustworthiness assessment checklist, and organizational risk assessment. We suggest that the evaluation is computationally carried out for trustworthiness recommendations, such as traceability, explainability, and accountability, in addition to traditional performance metrics such as novelty, quality, and reliability. © 2025 L. Siddharth and Jianxi Luo.

literature, feasibility is a basic evaluation that is needed before concepts can be judged for creativity, which includes novelty and value.[96] While these criteria do not assess trustworthiness, domain- or organization-specific quality control and risk assessment methods (e.g., failure modes and effect analysis) enable assessing quality, resiliency, and reliability, which are key recommendations for trustworthiness (refer to Table 1).

Model-based systems engineering is a widely adopted strategy in industries such as aerospace (e.g., Airbus), automotive (e.g., Ford Motor Company), and defense (e.g., Lockheed Martin) to structure the knowledge of artifacts in sufficient detail to enable risk assessment. As we proposed to generate concepts through RAG involving knowledge graph representations of prior knowledge, as depicted in Figure 3, the knowledge of concepts is already available in explicit forms that can serve as inputs for model-based system engineering methods—thereby enabling assessment of quality, resiliency, and reliability by considering design requirements (derived from opportunity statements in Section: Evaluate Opportunities for Trustworthy AI) and the organization-specific risk assessment strategies, as indicated in Figure 3.

While it is common for quality control and risk assessment to be carried out, trustworthy AI requires additional assessment criteria. Recalling the 1972 explosion of the Ford Pinto and the failure of electric car software in Volkswagen, scholars argue that the problems in these incidents arose not only from engineering failures but also from a lack of transparency and accountability.[97] Jana Fehr et al. recognize the lack of transparency in medical AI and evaluate 14 EU-based radiology products through a survey to find 29.1% median transparency.[98] Díaz-Rodríguez et al. note that the companies and individuals that develop AI models must be subject to audit for verification of traits such as stability, sensitivity, relevance, and reachability.[99] As systems are always subject to risks of different levels, it is necessary to be transparent about such risks, as recommended for AI-centric artifacts to be trustworthy AI. Such transparency can primarily be achieved through detailed, explainable, and traceable knowledge of concepts that form the AI-centric artifacts.

Assessing Artifacts for Trustworthy AI

In the previous sections, to contribute to design and innovation research, education, and practice, we leveraged the processes and underlying methods of the double hump model of data-driven innovation to explain how they can help operationalize generic, institutional recommendations for trustworthy AI. In this section, to contribute to technology policy and management, we briefly provide a data-driven innovation perspective on how existing AI-centric artifacts, their organizations, and the technological domains could be computationally assessed for trustworthiness.

Existing Frameworks

Governmental initiatives have proposed schemes to categorize the risks posed by AI-centric artifacts. The AI Act,[100] agreed upon by the EU Parliament and the EU Council, classifies the risk associated with AI systems as prohibited, high, limited, and minimal. This classification can also be understood in

96 L. Siddharth et al., "Toward Automatically Assessing the Novelty of Engineering Design Solutions," *Journal of Computing and Information Science in Engineering* 20, no. 1 (2020): article no. 011001, https://doi.org/10.1115/1.4044318.

97 Singh and Singh, "Wasabi," 23, 25; Kaur et al., "Trustworthy Artificial Intelligence," 18.

98 Jana Fehr et al., "A Trustworthy AI Reality-Check: The Lack of Transparency of Artificial Intelligence Products in Healthcare," *Frontiers in Digital Health* 6 (February 2024): article no. 1267290, https://doi.org/10.3389/fdgth.2024.1267290.

99 Díaz-Rodríguez et al., "Connecting the Dots," 1.

100 EU Council, "High-Level Summary of the AI Act," EU Artificial Intelligence Act, February 27, 2024, updated on May 30, 2024, https://artificialintelligenceact.eu/high-level-summary/.

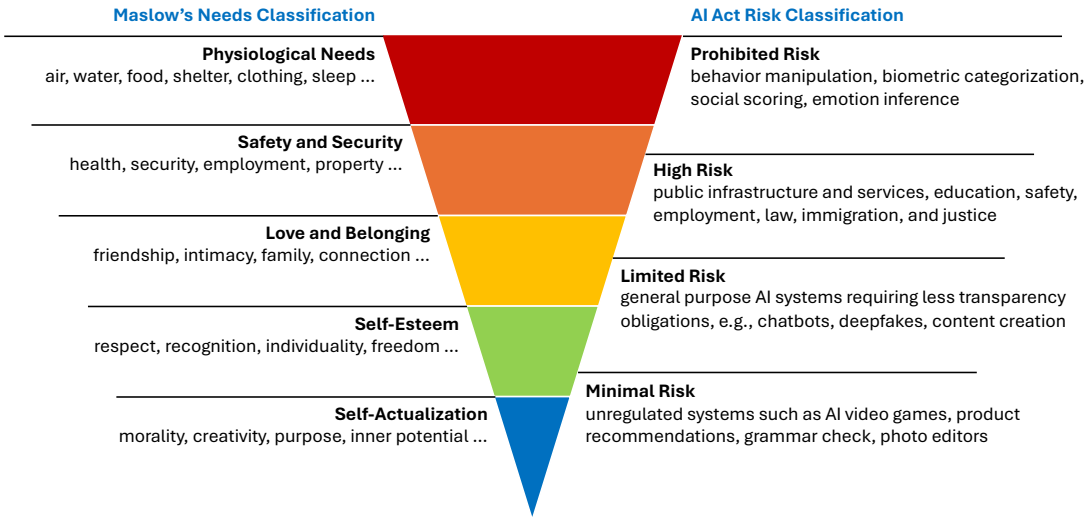

Figure 4
The EU AI Act classification of risk associated with AI systems understood in accordance with Maslow's Hierarchy of Needs. © 2025 L. Siddharth and Jianxi Luo.

accordance with Maslow's Hierarchy of Needs,[101] as depicted in Figure 4. Prohibited are the livelihood- and security-threatening systems (e.g., social scoring, biometric categorization) and manipulative systems (e.g., artificially inferring emotions for profitable and malicious intentions).

Under the high-risk category, the AI Act includes public infrastructure and services such as education, employment, law, migration, and justice. For the developers of high-risk AI, the AI Act mandates a risk management system, data governance, technical documentation, record keeping, usage instructions, human oversight, performance, and quality control. For systems with limited risk—like chatbots and content creators—the AI Act mandates technical documentation, copyright compliance, and usage instructions, while being transparent with the users that they are interacting with AI. The systems with minimal risk include tools for self-development, creativity-enhancing, and entertainment tools, among others.

In literature, Alfrink et al. classify domains such as child protection, public housing, health, social protection, security, and taxation under public AI and built environment and mobility solutions such as electric vehicle charging and parking systems under urban AI.[102] David Fernández-Llorca and Emilia Gómez review state-of-the-art autonomous vehicles in accordance with the recommendations for trustworthy AI (see Table 1). They find that "Technical Robustness & Safety" is *highly* mature, while other recommendations have *low* and *medium* levels of maturity.[103]

Potential Research Directions

Radclyffe et al. point out that generic means like the trustworthiness assessment checklist may not be compatible across organizations and domains that differ in size and levels of AI adoption.[104] There is a lack of technology and policy studies targeted at different technological domains and organizations for trustworthy implementations of AI-centric artifacts. Hence, the

101 Saul McLeod, "Maslow's Hierarchy of Needs," *Simply Psychology*, August 3, 2025, https://www.simplypsychology.org/maslow.html.
102 Alfrink et al., "Envisioning Contestability Loops," 55.
103 David Fernández-Llorca and Emilia Gómez, "Trustworthy Artificial Intelligence Requirements in the Autonomous Driving Domain," *Computer* 56, no. 2 (2023): 32, https://doi.org/10.1109/MC.2022.3212091.
104 Radclyffe et al., "Assessment List," 7.

literature is less cognizant of a variety of approaches with which existing artifacts could be assessed for trustworthiness. We therefore suggest the following research directions that could be pursued from a data-driven innovation standpoint.

Ontological basis of Trustworthy AI. The recommendations for trustworthy AI, such as private, robust, secure, fair, and traceable, are opinions from experts on the essential attributes of a trustworthy AI, which can be perceived differently by the users. In January 2021, after WhatsApp announced a privacy policy update that required sharing their data with Facebook, over 25 million users joined Telegram, and Signal saw 7.5 million downloads in a week. However, WhatsApp still dominates with around 3 billion users as of 2024.[105] Typically, users trust products and services in terms of brand reputation, transparency, customer service, consistent performance, word-of-mouth, and positive reviews, among others.

It is necessary to empirically understand whether the recommendations for trustworthy AI are, in principle, the indicators thereof. Design and innovation literature offers theoretical underpinnings of artifact-related constructs such as function, creativity, and quality. The construct function has continuously been theorized by engineering design scholars from different perspectives,[106] while also being empirically studied through various artifact descriptions. For instance, Jeremy Murphy et al. analyzed a corpus of over 65,000 patent descriptions and mined 1,700 verbs that represent function.[107] Likewise, through analyses of expert interviews, user surveys, case reports, and relevant documentation, indicators of trustworthiness could be gathered into an ontology of trustworthy AI comprising classes, subclasses, relationships, properties, definitions, and conditions.

Assessing Artifacts, Organizations, and Domains. Questionnaires and labelling instructions could be developed to create trustworthiness assessment datasets, leveraging a wide range of AI-centric artifacts from sources such as FutureTools.io,[108] There's an AI for That,[109] AITopTools,[110] and Product Hunt–AI Tools.[111] Evolving AI-centric prototypes of potential products and services can be identified through annual exhibitions (e.g., AI XPrize[112]) for startup trackers (e.g., Crunchbase[113]). Labelled datasets or some representative examples obtained from sources of AI-centric artifacts could be useful for training (multimodal) LLMs[114] for predicting trustworthiness scores from artifact descriptions.

In the technology management literature, patent quality is assessed using forward citations as well as the size of the patent domain,[115] as representations of patents include not only the artifact descriptions, but also inventors, organizations, domains, countries, and more.[116] The characterizations of domains and organizations rely on approaches such as knowledge trajectories, emerging technologies, and scientific engagement, which are mainly studied using citation and topic association networks.[117] Likewise, the trustworthiness of an AI-centric artifact could rely partly on its domains and organizations, which, on the other hand, require distinct approaches to be characterized for trustworthiness.

105 Business Insider report on the 2021 privacy policy update in Whatsapp, see Katie Canales, "WhatsApp Is Delaying a New Policy Change After Critics Claimed the Update Would Have Turned Over User Data to Facebook," Business Insider, January 16, 2021, https://www.businessinsider.com/whatsapp-privacy-policy-delay-three-months-2021-1.

106 Amaresh Chakrabarti et al., "A Case for Multiple Views of Function in Design Based on a Common Definition," *AI EDAM* 27, no. 3 (2013): 271–79, https://doi.org/10.1017/S0890060413000279; Nathan Crilly, "Function Propagation through Nested Systems," *Design Studies* 34, no. 2 (2013): 216–42, https://doi.org/10.1016/j.destud.2012.10.003; B. Chandrasekaran, "Representing Function: Relating Functional Representation and Functional Modeling Research Streams," *AI EDAM* 19, no. 2 (2005): 65–74, https://doi.org/10.1017/S0890060405050079.

107 Jeremy Murphy et al., "Function Based Design-by-Analogy: A Functional Vector Approach to Analogical Search," *Journal of Mechanical Design* 136, no. 10 (2014): article no. 101102, https://doi.org/10.1115/1.4028093.

108 FutureTools.io, https://www.futuretools.io/.

109 There's an AI for That, https://theresanaiforthat.com/.

110 AITopTools, https://aitoptools.com/.

111 Product Hunt — AI Tools, https://www.producthunt.com/topics/artificial-intelligence.

112 AI XPrize, https://www.xprize.org/prizes/artificial-intelligence.

113 Crunchbase, https://www.crunchbase.com/.

114 Jean-Baptiste Alayrac et al., "Flamingo: A Visual Language Model for Few-Shot Learning," in *NIPS '22: Proceedings of the 36th International Conference on Neural Information Processing Systems* (New York: ACM, 2022), article no. 1723, https://dl.acm.org/doi/10.5555/3600270.3601993.

115 Taiye Luo et al., "Multi-network Embeddedness and Innovation Quality: A Panel Data Analysis Based on fsQCA," *Journal of Engineering and Technology Management* 75 (January–March 2025): article no. 101860, https://doi.org/10.1016/j.jengtecman.2025.101860.

116 Guangtong Li et al., "Embedding Knowledge Graph of Patent Metadata to Measure Knowledge Proximity," *Journal of the Association for Information Science and Technology* 74, no. 4 (2023): 476–90, https://doi.org/10.1002/asi.24736.

Conclusions

The concerns regarding the trustworthiness of rapidly growing AI-centric artifacts have led to global recommendations from governmental, institutional, and scholarly initiatives. These recommendations (e.g., robust, private, secure) have been implemented in the form of assessment checklists that can be used to assess AI-centric artifacts retrospectively. However, these recommendations are still generic and have yet to be operationalized and integrated into the design and innovation process. To address this gap, we leveraged the double hump model of data-driven innovation, which integrates some transformative research in design and innovation literature in the last decade.

The double hump model of data-driven innovation provides a structured framework for leveraging digital footprints of design and innovations through advanced computational methods to enhance creativity and efficiency in the design and innovation process. By leveraging the model, we propose,

- Discovering opportunities for trustworthy AI from real-world AI-related incidents reported in news articles and legal case documents, to enable critical performances in terms of robustness, privacy, and safety.
- Evaluating opportunities for trustworthy AI upon generating contextualized opportunity statements that allude to the items in trustworthiness assessment checklists, to incorporate diversity, fairness, explainability, and other institutional recommendations.
- Generating concepts for trustworthy AI through RAG approaches, harnessing prior knowledge from patent documents and scientific articles, to enable traceability and explainability of concepts that form the AI-centric artifact.
- Evaluating concepts for trustworthy AI following organization- and domain-specific risk assessment strategies to assess quality, resiliency, and reliability, while also disclosing potential risks, for transparency, accountability, and auditability.

In technology and policy, the EU AI Act classifies AI systems according to the potential risk posed to social and environmental well-being. To build upon this classification and to carry out fine-grained assessments of AI-centric artifacts on trustworthiness, we propose developing an ontological basis of trustworthy AI, which could then lead to computational methods capable of assessing trustworthiness from the knowledge of AI-centric artifacts, their domains, and organizations.

Our future work is expected to address various shortcomings. Our propositions rely heavily on the availability and quality of structured data (e.g., patents, news, legal cases), which may not always be accessible across domains. The double hump model should be demonstrated holistically using case studies on the design of AI-centric artifacts, preferably through interdisciplinary design workshops. The double hump model should also integrate elements from other models of design and innovation (e.g., Stanford Design Thinking),[118] while also engaging AI development frameworks.

117 Laura Borge et al., "Assessing Interdisciplinary Research within an Emerging Technology Network: A Novel Approach Based on Patents in the Field of Bioplastics," *IEEE Transactions on Engineering Management* 71 (2024): 1452–69, https://doi.org/10.1109/TEM.2022.3146199; Zhiguo Cai et al., "Patent Analysis for Identifying Core Technology and Forecasting Promising Technology in Medical Imaging Equipment," *IEEE Transactions on Engineering Management* 71 (2024): 14375–86, https://doi.org/10.1109/TEM.2024.3453652.
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Declaration of Interest

There are no conflicts of interest involved in this article.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

While preparing this work, the authors used Grammarly and the built-in spell-grammar checker of Microsoft Word to support grammar and vocabulary review. All content was reviewed by the authors, who take full responsibility for the final published content.

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